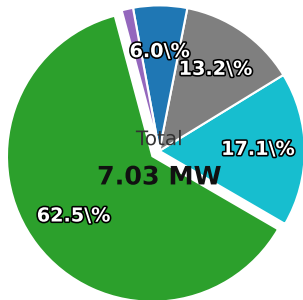


Single-pass ERL — wall-plug power budget
(10 GeV, 60 mA, $f_{\text{RF}} = 801.58$ MHz)



- SRF cryoplant (714 cavities, 2 K) — 4389 kW
- Injector RF ($\eta = 35\%$) — 1202 kW
- Residual RF (beam-loading) — 926 kW
- Beam dump (7 MeV $\times$ 60 mA) — 421 kW
- SR in return arc — 89 kW